178062: foliated biotite monzogranite, Wizard Bore

Location and sampling

SANDSTONE (SG 50-16), SANDSTONE (2742)
MGA Zone 50, 730380E 6929130N

Sampled on 30 October 2002

The sample was taken from a block, about 1 m high, 1 m wide and 2 m long, in an area of boulders 1.5 km south-southwest of Wizard Bore.

Tectonic unit/relations

The sample is from a light-grey, medium-grained, strongly foliated porphyritic biotite monzogranite within the Edale Shear Zone of the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Chen, 2005). The monzogranite contains pinkish feldspar phenocrysts, up to 3 cm long and aligned with the foliation. The sample was collected to determine the age of deformed rocks in the shear zone.

Petrographic description

The principal minerals are plagioclase (35 vol.%), microcline (30–35 vol.%), quartz (30 vol.%), and biotite (2–3 vol.%), with trace amounts of accessory opaque oxide, titanite, altered allanite, and zircon, and trace amounts of secondary chlorite, epidote, muscovite, and clay–sericite. Quartz occurs largely in anastomosing lamellae from 0.4 to 2 mm in width, as a mosaic of grains up to 1.5 mm in diameter, but there is no obvious quartz fabric. The plagioclase (?oligoclase) is partly as grains up to 4 mm long, whereas the largest microcline grains are as much as 8 mm long, with inclusions of quartz, plagioclase, and biotite. All the feldspar is anhedral. Minor myrmekite is present between the plagioclase and microcline. There are two types of lenses of recrystallized feldspar: one with grains about 0.4 mm in diameter, and the other has finer grains. Microcline and myrmekite are more abundant in the finer grained lenses. The biotite is fine grained, but some flakes are over 1 mm long. An anastomosing foliation, parallel to the quartz lenses, is present, but there is some poorly oriented biotite in pressure shadows and there could be an incipient S–C fabric. Oxide and rare altered allanite, up to 0.8 mm in grain size, are present, with rare titanite largely altered to leucoxene. Some plagioclase and microcline is deformed with bent twin planes, but quartz and finer grained feldspar are recrystallized. Minor muscovite has formed in and adjacent to biotite, with minor chlorite and rare epidote replacing the biotite. There is clay–sericite alteration in plagioclase and small areas within larger microcline grains. Epidote rims some of the altered allanite, and leucoxene has replaced titanite. The fabric shows evidence of deformation and metamorphism, probably initiated under amphibolite-facies conditions, with alteration continuing to low temperatures.

Zircon morphology

The zircons isolated from this sample are typically pale yellow or dark brown, elongate and irregular in shape, and between 50 × 80 µm and 200 × 320 µm in size. A majority of grains have faint internal zonation. Many grains have a speckled appearance in transmitted light, and many are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 29 May and 18 August 2003. The counter deadtime during both analysis sessions was 24 ns. During the first analysis sesssion, nine analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.33% (1σ). Analyses 1.1 to 9.1 were obtained during the first analysis session. During the second analysis session, five analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 3.04% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 1.1, 4.1, 5.1, 6.1, 6.2, 9.1, 9.2, 10.1, 12.3, 13.1, and 17.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-three analyses were obtained from 17 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with both recent and ancient episodes of radiogenic-Pb loss. Seventeen concordant to highly discordant analyses of 12 zircons (1.1, 3.1, 6.1, 6.2, 7.2, 8.1, 8.2, 9.1, 9.2, 10.1, 12.1, 12.2, 12.3, 13.1, 15.1, 16.1, 17.1) have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2697 ± 6 Ma (chi-squared = 0.66). Discordant analysis 14.1 indicates a significantly older $^{207}Pb/^{206}Pb$ date than the main population. The remaining five analyses are discordant (7.1), highly discordant (2.1, 5.1, 11.1), or highly reverse discordant (4.1).

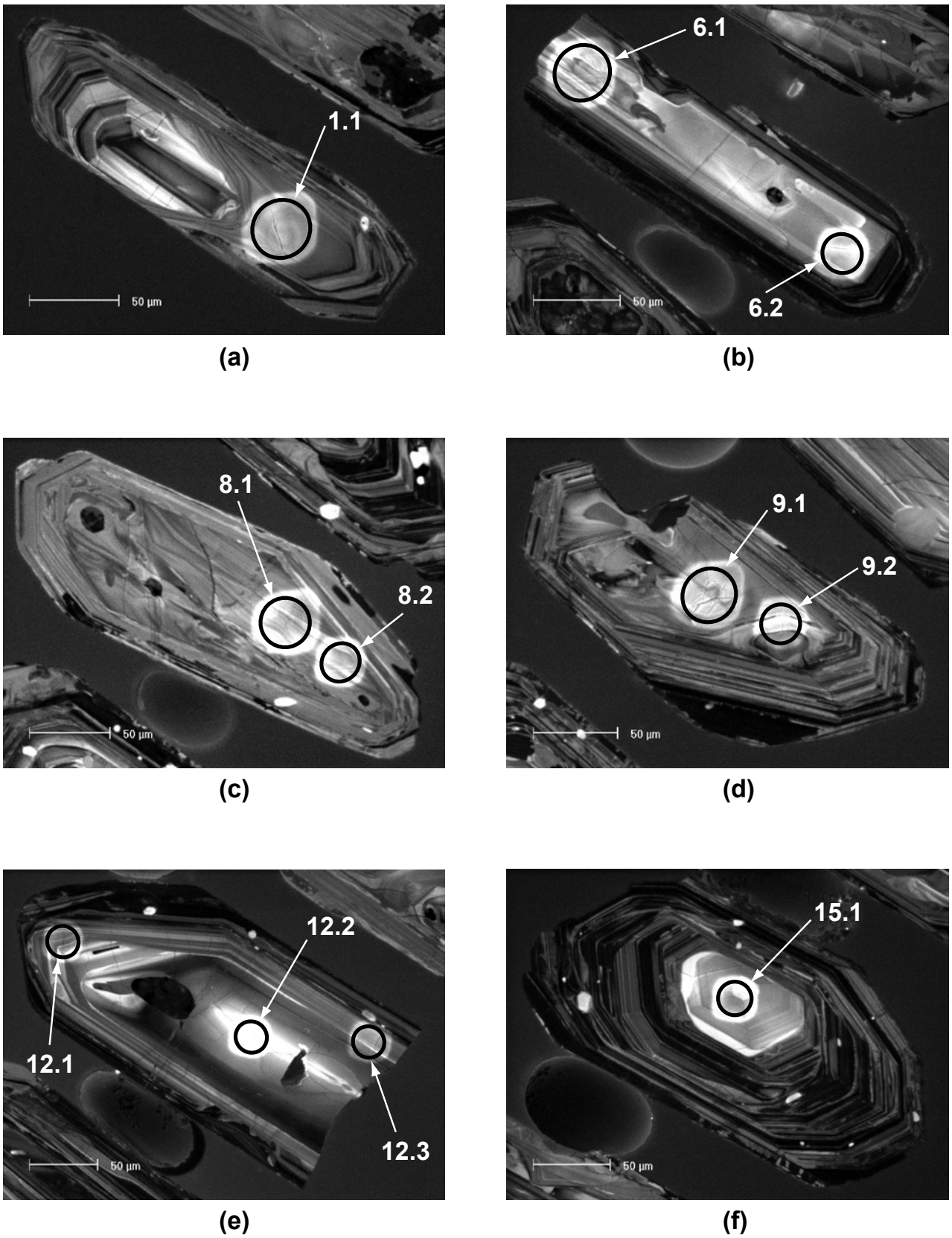

Figure 1. Cathodoluminescence images of representative zircons from sample 178062: foliated biotite monzogranite, Wizard Bore

Table 1. Ion microprobe analytical results for zircons from sample 178062: foliated biotite monzogranite, Wizard Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	207	274	114	0.684	0.18464	0.00081	0.19591	0.00150	0.4547	0.0062	11.575	0.171	90	2 695	7
2.1	273	155	138	0.219	0.17811	0.00061	0.17710	0.00101	0.4303	0.0058	10.566	0.151	88	2 635	6
3.1	162	102	94	0.711	0.18542	0.00095	0.19326	0.00179	0.4771	0.0065	12.197	0.185	93	2 702	8
4.1	602	407	749	0.715	0.19112	0.00039	0.81801	0.00112	0.7058	0.0095	18.599	0.257	125	2 752	3
5.1	421	260	161	3.523	0.17792	0.00139	0.42987	0.00315	0.2495	0.0034	6.119	0.100	55	2 634	13
6.1	99	211	70	1.832	0.18420	0.00157	0.40874	0.00350	0.4911	0.0068	12.473	0.215	96	2 691	14
7.1	626	109	320	0.224	0.18054	0.00037	0.04995	0.00048	0.4799	0.0064	11.945	0.165	95	2 658	3
8.1	169	110	101	0.216	0.18450	0.00074	0.17339	0.00120	0.5079	0.0069	12.921	0.190	98	2 694	7
9.1	147	202	95	1.580	0.18595	0.00119	0.26970	0.00245	0.4886	0.0067	12.528	0.198	95	2 707	11
10.1	254	171	162	4.214	0.18476	0.00281	0.40859	0.00630	0.4117	0.0126	10.489	0.377	82	2 696	25
11.1	423	398	232	0.877	0.16800	0.00114	0.37980	0.00257	0.4018	0.0123	9.306	0.299	86	2 538	11
12.1	170	187	116	0.346	0.18747	0.00139	0.30339	0.00270	0.5261	0.0162	13.598	0.443	100	2 720	12
13.1	196	149	116	3.351	0.18471	0.00277	0.27494	0.00593	0.4229	0.0130	10.771	0.387	84	2 696	25
14.1	72	48	53	0.468	0.23770	0.00251	0.17886	0.00420	0.5915	0.0186	19.386	0.668	96	3 105	17
15.1	69	131	54	0.717	0.18753	0.00280	0.52474	0.00672	0.5191	0.0163	13.423	0.490	99	2 721	25
16.1	124	182	72	1.294	0.18372	0.00252	0.52333	0.00606	0.3852	0.0119	9.758	0.345	78	2 687	23
17.1	236	366	145	1.407	0.18279	0.00175	0.39169	0.00385	0.4358	0.0134	10.984	0.365	87	2 678	16
6.2	81	171	68	3.882	0.18527	0.00408	0.45526	0.00925	0.5306	0.0167	13.553	0.550	102	2 701	36
8.2	195	143	118	0.816	0.18415	0.00162	0.21628	0.00311	0.4885	0.0150	12.404	0.410	95	2 691	15
9.2	228	290	148	5.982	0.18564	0.00356	0.51746	0.00819	0.3769	0.0116	9.646	0.369	76	2 704	32
7.2	664	114	360	0.202	0.18450	0.00075	0.05182	0.00090	0.5065	0.0154	12.885	0.403	98	2 694	7
12.2	306	168	45	6.466	0.17496	0.00638	0.59857	0.01503	0.0824	0.0025	1.988	0.101	20	2 606	61
12.3	252	345	117	2.926	0.18655	0.00266	0.67845	0.00653	0.2679	0.0082	6.892	0.244	56	2 712	23

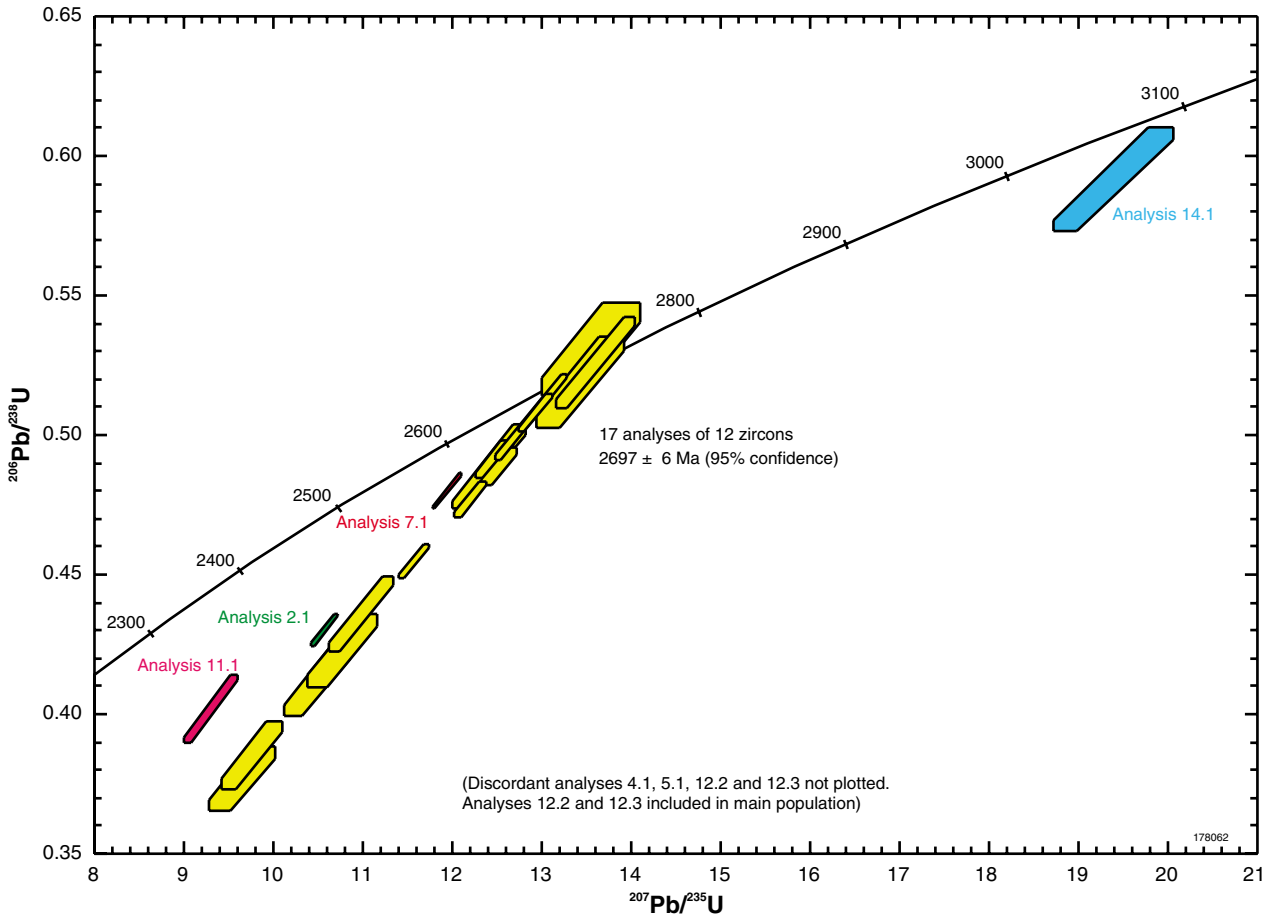

Figure 2. Concordia plot for sample 178062: foliated biotite monzogranite, Wizard Bore

The date of 2697 ± 6 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 17 concordant to highly discordant analyses of 12 zircons is interpreted as the age of igneous crystallization of the monzogranite. Discordant analysis 14.1 is interpreted to be of a xenocryst zircon. The remaining analyses are interpreted to be of sites that underwent redistribution of radiogenic-Pb during ancient and recent disturbance episodes.

Recommended reference for this publication:

NELSON, D. R., 2005, 178062: foliated biotite monzogranite, Waukenjerrie Well; Geochronology dataset 589; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 18/08/2003; Data released: 30/06/2005